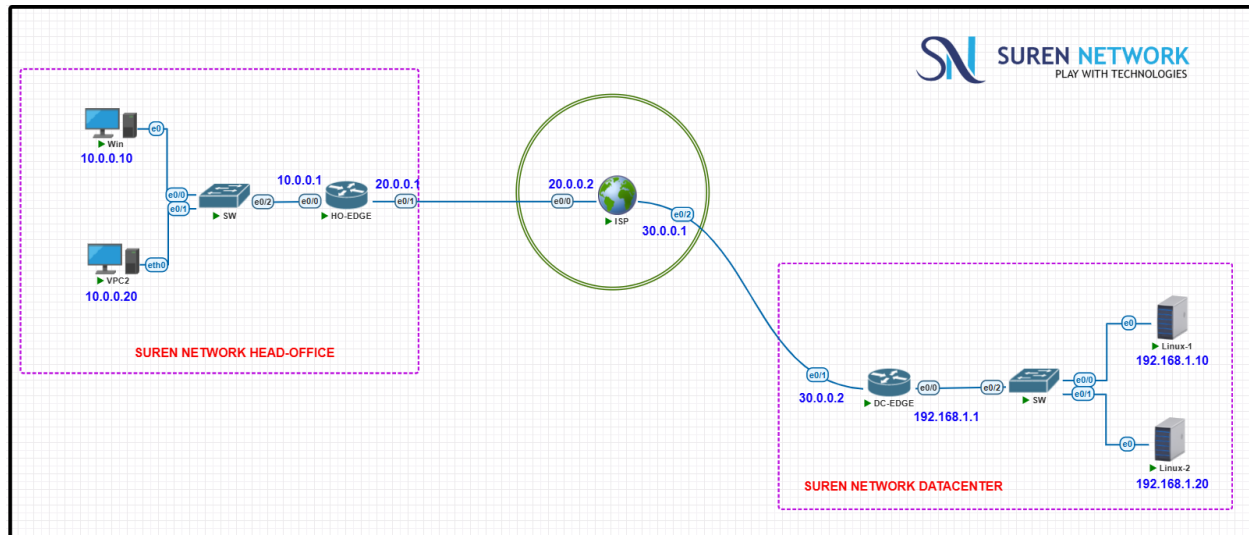


GRE VPN



Create default route in head office router

```
HO-EDGE
HO-EDGE(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
HO-EDGE(config)#
HO-EDGE(config)#
```

Create gre vpn configuration in head office router

```
HO-EDGE
HO-EDGE(config)#interface tunnel 1
HO-EDGE(config-if)#ip a
*Sep 24 03:15:19.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, c
changed state to down
HO-EDGE(config-if)#ip add 1.0.0.1 255.0.0.0
HO-EDGE(config-if)#tunnel source 20.0.0.1
HO-EDGE(config-if)#tunnel destination 30.0.0.2
HO-EDGE(config-if)#
*Sep 24 03:16:22.232: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, c
changed state to up
HO-EDGE(config-if)#exit
HO-EDGE(config)#
```

Create vpn route in head office router

```
HO-EDGE
HO-EDGE(config)#
HO-EDGE(config)#
HO-EDGE(config)#ip route 192.168.1.0 255.255.255.0 tunnel 1
HO-EDGE(config)#
```

Create default route in datacenter router

```
DC-EDGE
DC-EDGE(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
DC-EDGE(config)#
DC-EDGE(config)#
```

Create gre vpn configuration in datacenter router

```
DC-EDGE
DC-EDGE(config)#interface tunnel 1
DC-EDGE(config-if)#ip a
*Sep 24 03:23:09.715: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, c
hanged state to down
DC-EDGE(config-if)#ip add 1.0.0.2 255.0.0.0
DC-EDGE(config-if)#tunnel source 30.0.0.2
DC-EDGE(config-if)#tunnel destination 20.0.0.1
DC-EDGE(config-if)#ex
*Sep 24 03:23:53.929: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, c
hanged state to up
DC-EDGE(config-if)#exit
DC-EDGE(config)#
```

Create vpn route in datacenter router

```
DC-EDGE
DC-EDGE(config)#
DC-EDGE(config)#ip route 10.0.0.0 255.0.0.0 tunnel 1
DC-EDGE(config)#
DC-EDGE(config)#
```

Check gre tunnel status

```
HO-EDGE
HO-EDGE(config)#do sh int tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
    Internet address is 1.0.0.1/8
    MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
      reliability 255/255, txload 1/255, rxload 1/255
    Encapsulation TUNNEL, loopback not set
    Keepalive not set
    Tunnel linestate evaluation up
    Tunnel source 20.0.0.1, destination 30.0.0.2
    Tunnel protocol/transport GRE/IP
      Key disabled, sequencing disabled
      Checksumming of packets disabled
    Tunnel TTL 255, Fast tunneling enabled
    Tunnel transport MTU 1476 bytes
    Tunnel transmit bandwidth 8000 (kbps)
    Tunnel receive bandwidth 8000 (kbps)
    Last input 00:18:30, output never, output hang never
    Last clearing of "show interface" counters 00:42:59
    Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
    Queueing strategy: fifo
    Output queue: 0/0 (size/max)
    5 minute input rate 0 bits/sec, 0 packets/sec
    5 minute output rate 0 bits/sec, 0 packets/sec
      22 packets input, 1992 bytes, 0 no buffer
      Received 0 broadcasts (0 IP multicasts)
      0 runs, 0 giants, 0 throttles
      0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
      12 packets output, 1008 bytes, 0 underruns
      0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
      0 output buffer failures, 0 output buffers swapped out
HO-EDGE(config)#

DC-EDGE
DC-EDGE(config)#do sh int tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
    Internet address is 1.0.0.2/8
    MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
      reliability 255/255, txload 1/255, rxload 1/255
    Encapsulation TUNNEL, loopback not set
    Keepalive not set
    Tunnel linestate evaluation up
    Tunnel source 30.0.0.2, destination 20.0.0.1
    Tunnel protocol/transport GRE/IP
      Key disabled, sequencing disabled
      Checksumming of packets disabled
    Tunnel TTL 255, Fast tunneling enabled
    Tunnel transport MTU 1476 bytes
    Tunnel transmit bandwidth 8000 (kbps)
    Tunnel receive bandwidth 8000 (kbps)
    Last input 00:14:49, output 00:14:49, output hang never
    Last clearing of "show interface" counters 00:36:04
    Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
    Queueing strategy: fifo
    Output queue: 0/0 (size/max)
    5 minute input rate 0 bits/sec, 0 packets/sec
    5 minute output rate 0 bits/sec, 0 packets/sec
      12 packets input, 1008 bytes, 0 no buffer
      Received 0 broadcasts (0 IP multicasts)
      0 runs, 0 giants, 0 throttles
      0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
      22 packets output, 1992 bytes, 0 underruns
      0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
      0 output buffer failures, 0 output buffers swapped out
DC-EDGE(config)#
```

Check reachability from ho private network to datacenter private network

```
QEMU (Win)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=3ms TTL=62
Reply from 192.168.1.10: bytes=32 time=3ms TTL=62
Reply from 192.168.1.10: bytes=32 time=3ms TTL=62
Reply from 192.168.1.10: bytes=32 time=3ms TTL=62

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms
C:\Documents and Settings\user>color f0

QEMU (Linux-1)
PING 10.0.0.10 (10.0.0.10): 56 data bytes
64 bytes from 10.0.0.10: seq=0 ttl=126 time=4.212 ms
64 bytes from 10.0.0.10: seq=1 ttl=126 time=4.230 ms
64 bytes from 10.0.0.10: seq=2 ttl=126 time=4.759 ms
64 bytes from 10.0.0.10: seq=3 ttl=126 time=3.558 ms
64 bytes from 10.0.0.10: seq=4 ttl=126 time=4.129 ms
64 bytes from 10.0.0.10: seq=5 ttl=126 time=4.050 ms
64 bytes from 10.0.0.10: seq=6 ttl=126 time=4.704 ms
64 bytes from 10.0.0.10: seq=7 ttl=126 time=2.904 ms
64 bytes from 10.0.0.10: seq=8 ttl=126 time=3.521 ms
64 bytes from 10.0.0.10: seq=9 ttl=126 time=3.234 ms
64 bytes from 10.0.0.10: seq=10 ttl=126 time=4.054 ms
64 bytes from 10.0.0.10: seq=11 ttl=126 time=3.790 ms
64 bytes from 10.0.0.10: seq=12 ttl=126 time=4.367 ms
64 bytes from 10.0.0.10: seq=13 ttl=126 time=2.770 ms
64 bytes from 10.0.0.10: seq=14 ttl=126 time=3.939 ms
64 bytes from 10.0.0.10: seq=15 ttl=126 time=4.494 ms
64 bytes from 10.0.0.10: seq=16 ttl=126 time=3.917 ms
64 bytes from 10.0.0.10: seq=17 ttl=126 time=4.228 ms
64 bytes from 10.0.0.10: seq=18 ttl=126 time=3.427 ms

--- 10.0.0.10 ping statistics ---
19 packets transmitted, 19 packets received, 0% packet loss
round-trip min/avg/max = 2.770/3.909/4.759 ms
```